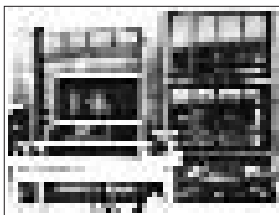


1.10 Wartime 1943

January 1943

The Harvard Mark I, which was originally called the ASCC Mark I, Harvard-IBM Automatic Sequence Controlled Calculator, was built at Harvard University by Howard Aiken and his team. The project was partly financed by IBM. It is said that it became the first program-controlled calculator, though as we mentioned above, the Z3 lays claim to that, too. The machine weighed five tonnes.



The Z3 is the Desktop-like thing at the left; this is an actual photograph

The program was read from a paper tape; data could be read from other tapes or card readers or the constant registers (refer chapter 2 for an explanation of registers). Later, the Mark I was modified to support multiple paper tape readers for the program. The transfer of the running program from one of the tapes to the other could be conditional, which meant it supported a conditional subroutine call. (Non-programmers, skip!)

On maxmon.com is this interesting snippet: “Many consider that the modern computer era commenced with the first large-scale automatic digital computer, which was developed between 1939 and 1944 (see also Konrad Zuse and his Z3 computer).” This underscores the fact that both are strong contenders to the throne of “first computer,” but the idea is, the Mark I was large-scale.



Howard Aiken, the computer icon that side of the Atlantic

Visit <http://tinyurl.com/2w4zav> for IBM’s perspective of the Mark I.

Sometime later in 1943 began Project Whirlwind, to design a flight simulator for bombing crews. It was completed in the 50s.